JOBSHEET 15

Collection

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13.2) Persiapan.

Code Customer.java:

```java
package Jobsheet15;

public class Customer {
    public int id;
    public String name;

    public Customer() {
    }

    public Customer(int id, String name) {
        this.id = id;
        this.name = name;
    }

    public String toString() {
        return " ID: " + this.id + " Nama: " + this.name;
    }

}
```

Code book.java:

```java
package Jobsheet15;

public class Book {
    public String isbn;
    public String title;

    public Book() {

    }

    public Book(String isbn, String title) {
        this.isbn = isbn;
        this.title = title;
    }

    public String toString() {
        return "ISBN: " + this.isbn + " Title: " + this.title;
    }

}
```

13.3) Praktikum (Implementasi ArrayList).

Code DemoArrayList.java:

```
Customer.java U    Book.java U    DemoArrayList.java U ×
Jobsheet15 > DemoArrayList.java > DemoArrayList
package Jobsheet15;

import java.util.ArrayList;

public class DemoArrayList {
    Run | Debug
    public static void main(String[] args) {
        ArrayList<Customer> customers = new ArrayList<>(initialCapacity: 0);
        Customer customer1 = new Customer(id: 1, name: "Zakia");
        Customer customer2 = new Customer(id: 5, name: "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new Customer(id: 4, name: "Cicak"));

        customers.add(index: 2, new Customer(id: 100, name: "Rosa"));

        System.out.println(customers.indexOf(customer2));
        System.out.println();

        Customer customer = customers.get(index: 1);
        System.out.println(customer.name);
        customer.name = "Budi Oetomo";

        ArrayList<Customer> newCustomers = new ArrayList<>();
        newCustomers.add(new Customer(id: 201, name: "Della"));
        newCustomers.add(new Customer(id: 202, name: "Victor"));
        newCustomers.add(new Customer(id: 203, name: "Sarah"));

        customers.addAll(newCustomers);

        System.out.println(customers);

        for (Customer cust : customers) {
            System.out.println(cust.toString());
        }
    }
}
```

Outputnya:

```
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\wo
itmaStrukturData_b46f464f\bin' 'Jobsheet15.DemoArrayList'
1

Budi
[ ID: 1 Nama: Zakia,  ID: 5 Nama: Budi Oetomo,  ID: 100 Nama: Rosa,  ID: 4 Nama: Cicak,  ID: 201 Nama: Della,  ID: 202 Nama: Victor,  ID: 203 Nama: Sarah]
 ID: 1 Nama: Zakia
 ID: 5 Nama: Budi Oetomo
 ID: 100 Nama: Rosa
 ID: 4 Nama: Cicak
 ID: 201 Nama: Della
 ID: 202 Nama: Victor
 ID: 203 Nama: Sarah
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

Langkah No.5

Compile dan run kode program, di mana object yang baru ditambahkan? Di awal, di tengah, atau di akhir collection?

Jawaban: Penambahan data akan di tempatkan di akhir collection

Langkah No.7

Compile dan run kode program. Index pada ArrayList dimulai dari 0 atau 1?

Jawaban: Index pada array list dimulai dari 0, dibuktikan dari ketika di print customers.indexOf(customer2) yang keluar adalah 1

Langkah No.10

Cobalah hapus angka 2 saat instansiasi object customers. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

Ya,ArrayList bisa di ininstansiasi tanpa adanya size awal

13.4) Praktikum (Implementasi Stack).

Code StackDemo.java:

```
Customer.java U    Book.java U    DemoArrayList.java U    StackDemo.java U
Jobsheet15 > StackDemo.java > StackDemo
package Jobsheet15;

import java.util.Stack;

public class StackDemo {
    Run | Debug
    public static void main(String[] args) {
        Book book1 = new Book(isbn: "1234", title: "Dasar Pemrograman");
        Book book2 = new Book(isbn: "7145", title: "Hafalan Sholat Delisa");
        Book book3 = new Book(isbn: "3562", title: "Muhammad Al-Fatih");

        Stack<Book> books = new Stack<>();
        books.push(book1);
        books.push(book2);
        books.push(book3);

        Book temp = books.peek();
        if (temp != null) {
            System.out.println(temp.toString());
        }
        Book temp2 = books.pop();

        if (temp2 != null) {
            System.out.println(temp2.toString());
        }
        for (Book book : books) {
            System.out.println(book.toString());
        }

        int posisi = books.search(o: "Dasar Pemrograman");
        System.out.println("Posisi " + posisi);
    }
}
```

Outputnya:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    GITLENS

PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCo
ws\PraktikumAlgoritmaStrukturData_b46f464f\bin' 'Jobsheet15.StackDemo'
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar Pemrograman
ISBN: 7145 Title: Hafalan Sholat Delisa
Posisi -1
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

Langkah No.5

Mengapa perlu ada pengecekan (temp != null)?

Jawaban: Untuk melakukan pengecekan bahwa objek temp tidak kosong untuk melakukan fungsi toString.

Langkah No.8

Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

Jawaban: int posisi = books.search("Dasar Pemrograman");

System.out.println("Posisi " + posisi);

13.5) Praktikum (Implementasi TreeSet).

Code TreeSetDemo.java:

```
Customer.java U    Book.java U    DemoArrayList.java U    StackDemo.java U    TreeSetDemo.java U
Jobsheet15 > TreeSetDemo.java > TreeSetDemo

package Jobsheet15;

import java.util.TreeSet;

public class TreeSetDemo {
    Run | Debug
    public static void main(String[] args) {
        TreeSet<String> fruits = new TreeSet<>();

        fruits.add(e: "Mangga");
        fruits.add(e: "Apel");
        fruits.add(e: "Jeruk");
        fruits.add(e: "Jambu");

        for (String temp : fruits){
            System.out.println(temp);
        }
        System.out.println("First: " + fruits.first());
        System.out.println("Last: " + fruits.last());

        fruits.remove(o: "Jeruk");
        System.out.println("Setelah Remove " + fruits);

        fruits.pollFirst();
        System.out.println("Setelah poll first " + fruits);

        fruits.pollLast();
        System.out.println("Setelah Poll last " + fruits);
    }
}
```

Outputnya:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   GITLENS

PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptio
ws\PraktikumAlgoritmaStrukturData_b46f464f\bin' 'Jobsheet15.TreeSetDemo'
Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah Remove [Apel, Jambu, Mangga]
Setelah poll first [Jambu, Mangga]
Setelah Poll last [Jambu]
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

Langkah No.4

Compile dan run program. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?

Jawaban: Karena dalam TreeSet urutan akan ada dalam bentuk alphabetical mulai dari a (apel), j (jambu, jeruk),m (mangga).

Langkah No.6

Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?

Jawaban: First() akan menampilkan data terdepan, last() akan menampilkan data terakhir, sedangkan pollFirst() akan mengambil data terdepan dan pollLast() akan mengambil data terakhir.

13.6) Praktikum (Sorting).

Code SortDemo.java:

```java
package Jobsheet15;

import java.util.ArrayList;
import java.util.Collections;

public class SortDemo {
    public static void main(String[] args) {
        ArrayList<String> daftarSiswa = new ArrayList<>();
        daftarSiswa.add(e: "Zainab");
        daftarSiswa.add(e: "Andi");
        daftarSiswa.add(e: "Rara");
        Collections.sort(daftarSiswa);

        System.out.println(daftarSiswa);
    }
}
```

Code DemoArrayList.java:

```java

        customers.sort((c1, c2) -> c1.name.compareTo(c2.name));
        System.out.println(customers);
    }
}
```

Outputnya:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   GITLENS

PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>  & 'C:\Program Files\Java\jdk-24\b
ws\PraktikumAlgoritmaStrukturData_b46f464f\bin' 'Jobsheet15.SortDemo'
[Andi, Rara, Zainab]
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```